Unit 2 Guided Notes — Exponent Rules

ANSWER KEY

Same base? Move the exponents.

🎯 By the end of this unit, I can…

- Apply the product, quotient, and power rules of exponents
- Simplify expressions with negative and zero exponents

Standards: NC.8.EE.1

Answer key: every blank and final answer is filled in green.

1 Exponent Rules

📖 Words to know

Base: the number being multiplied. In 2^5, the base is 2.

Exponent: the small number up high. It tells how many times to multiply the base. In 2^5, the exponent is 5.

Coefficient: the number in front of a variable. In $4x^3$, the coefficient is 4.

Monomial: one term: a number, a variable, or a product of them, like $4x^3$ or $-2xy^2$.

📌 **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Product Rule — Count the Factors

x^3 [x] [x] [x]

$\times x^2$ [x] [x]

$= x^5$

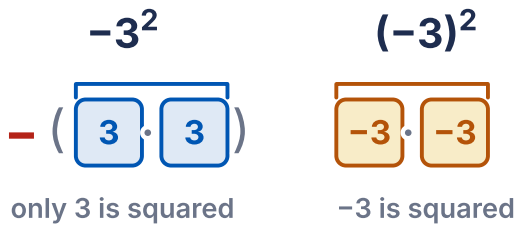
3 + 2 = 5 factors of x

$x^3 \times x^2 = x^{3+2} = x^5$. Count the factors: 3 + 2 = 5 → **ADD** the exponents.

The Five Core Rules

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	$x^3 \cdot x^4 = x^7$	$(2x^3)(3x^2) = 6x^5$	Same base? ADD
Quotient Rule	$x^a \div x^b = x^{a-b}$	$x^6 \div x^2 = x^4$	$(12x^7) \div (3x^2) = 4x^5$	Same base? SUBTRACT
Power Rule	$(x^a)^b = x^{a \cdot b}$ (multiply the exponents)	$(x^2)^3 = x^6$	$(2x^3)^2 = 4x^6$	Power to power? MULTIPLY
Zero Exponent	$x^0 = 1$	$5^0 = 1$	$(3x^5)^0 = 1$	ANY base^0=1
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	$2^{-3} = 1/8$	$4x^{-3} = 4/x^3$	Flip to fraction

⚠️ **PARENTHESES MATTER!** -3^2 = **−9** vs. $(-3)^2$ = **9**. Without ( ), ONLY the base gets the exponent — not the negative sign!

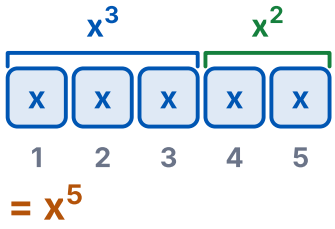


The little ^{2} belongs to whatever sits right next to it: the 3 alone, or the whole (−3).

Modeling the Rules — See WHY

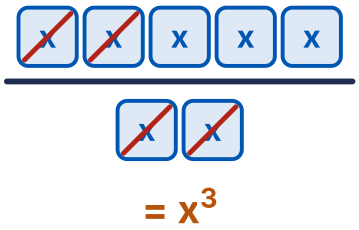
Each rule comes from writing out the repeated multiplication.

Product Rule · $x^3 \times x^2 = x^5$



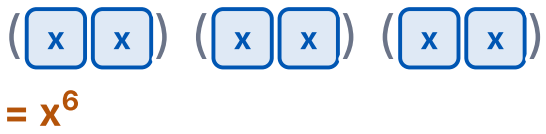
$x \cdot x \cdot x \times x \cdot x = x \cdot x \cdot x \cdot x \cdot x = x^5$. Count the factors: 3 + 2 = 5. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$



$x \cdot x \cdot x \cdot x \cdot x \div x \cdot x$ → cancel 2 matching pairs → $x \cdot x \cdot x = x^3$. Cancel matching factors: 5 − 2 = 3. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$



3 groups of 2 factors = 6 factors

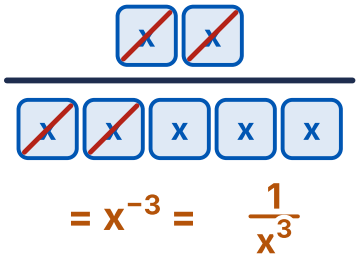
Three copies of x^2: $(x \cdot x)(x \cdot x)(x \cdot x) = x^6$. 2 × 3 = 6. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

$$2^3 = 8 \quad \div 2$$
$$2^2 = 4 \quad \div 2$$
$$2^1 = 2 \quad \div 2$$
$$2^0 = 1$$

Count down by 2s: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$. Each time the exponent drops by 1, divide by 2. So $2^0 = 2 \div 2 = 1$. This works for any base, so $x^0 = 1$.

Negative Exponent · $x^{-3} = \frac{1}{x^3}$



$x^2 \div x^5$: cancel 2 pairs of x. 3 x's are left on the bottom. So $x^{-3} = \frac{1}{x^3}$. A negative exponent means flip it to the bottom: $x^{-n} = \frac{1}{x^n}$.

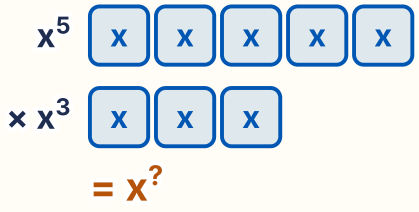
The Pattern Behind x^0 and x^{-n}

$$2^3 = 8 \quad \div 2$$
$$2^2 = 4 \quad \div 2$$
$$2^1 = 2 \quad \div 2$$
$$2^0 = 1 \quad \div 2$$
$$2^{-1} = \frac{1}{2} \quad \div 2$$
$$2^{-2} = \frac{1}{4}$$

Each step down the exponent drops by 1 and the value is **÷ 2**. Keep going past 2^1 and you get $2^0 = 1$, then fractions.

Quick Check

MC **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**



Same base, multiplied: count every x tile.

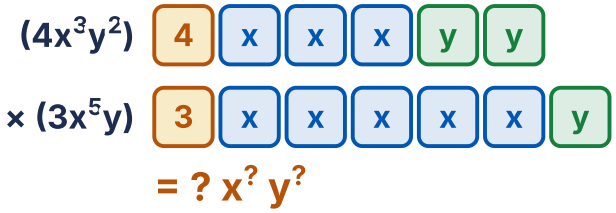
- ● **x^8 (Product Rule)**
- ○ x^{15} (Power Rule)
- ○ x^2 (Quotient Rule)

Match the Rule to its Key Phrase

MATCH **Write the letter of each rule's key phrase on the line.**

B	1. Product Rule	**A.**	Power to power? MULTIPLY
D	2. Quotient Rule	**B.**	Same base? ADD exponents
A	3. Power Rule	**C.**	Flip to a fraction
C	4. Negative Exponent	**D.**	Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$



Multiply the gold numbers. Count the blue x tiles and the green y tiles.

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

ERROR ANALYSIS Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?

$(3x^2)$ [3] [x] [x]

$\times (3x^2)$ [3] [x] [x]

$\times (3x^2)$ [3] [x] [x]

$(3x^2)^3$ means three copies of $3x^2$. Look at the gold tiles: how many 3s are there?

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Exponent Rules with Monomials

1) Simplify $x^3 \cdot x^5$

x^3 [x] [x] [x]

$\times x^5$ [x] [x] [x] [x] [x]

$= x^?$

Count all the x tiles.

Answer: x^8

2) Simplify $(y^4)^3$

y^4 [y] [y] [y] [y]

$\times y^4$ [y] [y] [y] [y]

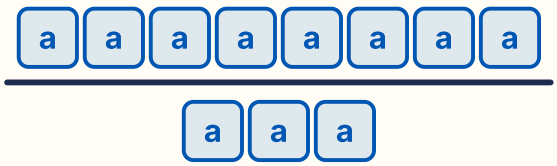
$\times y^4$ [y] [y] [y] [y]

$= y^?$

$(y^4)^3$ means 3 copies of y^4.

Answer: y^{12}

3) Simplify $a^8 \div a^3$

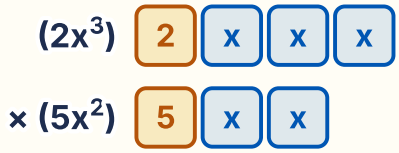


$= a^{?}$

Cancel matching pairs, top and bottom.

Answer: a^5

4) Simplify $(2x^3)(5x^2)$

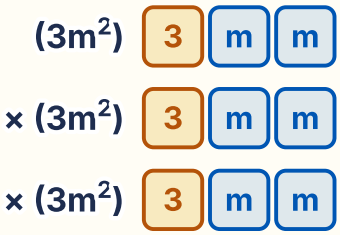


$= ? \; x^{?}$

Multiply the numbers; count the x tiles.

Answer: $10x^5$

5) Simplify $(3m^2)^3$

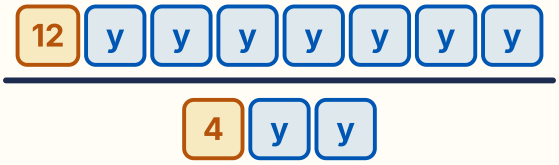


$= ? \; m^{?}$

3 copies: every copy has a 3 and two m tiles.

Answer: $27m^6$

6) Simplify $12y^7 \div 4y^2$



$= ? \; y^{?}$

Divide the numbers; cancel matching y pairs.

Answer: $3y^5$

7) Simplify $(-2a^4)^2$

$(-2a^4)$ [−2] [a] [a] [a] [a]

$\times (-2a^4)$ [−2] [a] [a] [a] [a]

$= ?\ a^{?}$

2 copies: $(-2)(-2)$ and all the a tiles.

Answer: $4a^8$

8) Simplify $(5x^2y)(3xy^3)$

$(5x^2y)$ [5] [x] [x] [y]

$\times (3xy^3)$ [3] [x] [y] [y] [y]

$= ?\ x^{?}\ y^{?}$

Multiply the numbers; count x tiles and y tiles.

Answer: $15x^3y^4$

9) Simplify $18n^6 \div 6n^6$

$$\frac{[18]\,[n]\,[n]\,[n]\,[n]\,[n]\,[n]}{[6]\,[n]\,[n]\,[n]\,[n]\,[n]\,[n]}$$

$= ?$

Divide the numbers; cancel matching n pairs.

Answer: 3

10) Simplify $(4b^3)^2 \div 2b$

$$\frac{([4]\,[b]\,[b]\,[b])\,([4]\,[b]\,[b]\,[b])}{[2]\,[b]}$$

$= ?\ b^{?}$

$(4b^3)^2$ is 2 copies of $4b^3$ on top.

Answer: $8b^5$

11) Simplify $(-3x^2)(-4x^3)$

$(-3x^2)$ [−3] [x] [x]

$\times (-4x^3)$ [−4] [x] [x] [x]

$= ? \; x^{?}$

Watch the signs: negative × negative.

Answer: $12x^5$

12) Simplify $(x^5y^2)^3$

(x^5y^2) [x] [x] [x] [x] [x] [y] [y]

$\times (x^5y^2)$ [x] [x] [x] [x] [x] [y] [y]

$\times (x^5y^2)$ [x] [x] [x] [x] [x] [y] [y]

$= x^{?} \; y^{?}$

3 copies of x^5y^2.

Answer: $x^{15}y^6$